Chapter 9, Problem 2PP

Problem

Find the phase angle between

$i_1 = -4\sin(377t + 55°)$ and $i_2 = 5\cos(377t - 65°)$

Does *i*1 lead or lag *i*2?

Step-by-step solution

Step 1 of 1

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Consider the signals $i_1 = -4\sin(377t + 55°)$ and $i_2 = 5\cos(377t - 65°)$.

Express the signal i_1 in standard form of a sinusoid.

$$\begin{aligned} i_1 &= -4\sin(377t + 55°) \\ &= 4\sin(377t + 55° + 180°) \quad [\text{since, } \sin(\omega t + 180°) = -\sin(\omega t)] \end{aligned}$$

$i_1 = 4\sin(377t + 235°)$ ...... (1)

Therefore, the alternate form of i_1 is $4\sin(377t + 235°)$.

Similarly express the signal i_2 in standard form of a sinusoid.

$$\begin{aligned} i_2 &= 5\cos(377t - 65°) \\ &= 5\sin(377t + 90° - 65°) \quad [\text{since, } \sin(\omega t + 90°) = \cos(\omega t)] \end{aligned}$$

$i_2 = 5\sin(377t + 25°)$ ...... (2)

Therefore, the alternate form of i_2 is $5\sin(377t + 25°)$.

Compare the equations (1) and (2) to find the phase difference.

$$\begin{aligned} \phi &= 235° - 25° \\ &= 210° \end{aligned}$$

Hence the phase difference is $\boxed{210°}$ and i_1 leads i_2.